Topological Neuroimaging: Alzheimer's Disease Classification via Interpretable Graph Attention Networks

Yuval Shah
MBA-TECH AI (Roll No: R062)
NMIMS MPSTME MUMBAI

Vedant Pandey
MBA-TECH AI (Roll No: R026)
NMIMS MPSTME MUMBAI

***Abstract*—The classification of Alzheimer's Disease (AD) from magnetic resonance imaging (MRI) has traditionally relied on high-dimensional convolutional neural networks or flat tabular machine learning baselines, both of which often lack structural interpretability. In this paper, we propose a novel framework utilizing Graph Attention Networks (GATs) to model the brain as a topological graph, leveraging volumetric tissue segmentation. Utilizing 161 clinical subjects from the OASIS-1 dataset, we construct 3-node micro-graphs representing Cerebrospinal Fluid (CSF), Gray Matter, and White Matter. Our GAT architecture achieved a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of 0.8360, significantly outperforming the standard Logistic Regression tabular baseline (ROC-AUC 0.7933) under strict 5-fold cross-validation. Furthermore, through an extraction of the network's attention weights, we demonstrate that the model independently localized disease pathology, assigning maximum predictive importance to CSF expansion (0.3965 mean attention weight), mathematically mirroring the clinical progression of hydrocephalus ex vacuo. An ablation study isolating tissue volume from demographic features confirmed the intrinsic predictive power of the graph topology (0.7200 ROC-AUC). These findings suggest that GNNs provide a highly accurate, computationally efficient, and clinically interpretable approach to neurodegenerative disease classification.**

***Index Terms*—Graph Attention Networks, Alzheimer's Disease, Neuroimaging, Deep Learning, Interpretability, Computational Complexity**

I. Introduction

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder characterized by a gradual decline in cognitive function and severe brain atrophy. Early and accurate diagnosis relies heavily on clinical assessments paired with structural Magnetic Resonance Imaging (MRI). As the global incidence of dementia rises, the demand for scalable, automated, and interpretable computer-aided diagnostic (CAD) systems has become paramount.

Historically, machine learning approaches for AD classification have diverged into two distinct methodologies. The first utilizes tabular clinical and demographic data (e.g., Age, Education, Total Intracranial Volume) fed into traditional classifiers such as Support Vector Machines (SVM) or Random Forests. While computationally lightweight, these models inherently discard the spatial and structural relationships within the brain. The second approach employs 3D Convolutional Neural Networks (CNNs) on raw MRI voxels. While often highly accurate, 3D CNNs are computationally expensive, prone to overfitting on small medical datasets, and function largely as "black boxes," making it difficult for clinicians to interpret the morphological basis of the prediction [1].

Graph Neural Networks (GNNs), and specifically Graph Attention Networks (GATs), offer a compelling geometric alternative [2]. By representing the brain as a topological graph—where regions of interest operate as nodes and their spatial or functional relationships as edges—we can preserve structural geometry while maintaining parameter efficiency.

In this study, we hypothesize that translating 3D MRI segmentations into topological micro-graphs will yield superior classification metrics compared to flat tabular models, while simultaneously allowing for robust clinical interpretability through attention weight analysis.

II. Related Work

The application of machine learning to neuroimaging for the diagnosis of Alzheimer's Disease has undergone a rapid evolution over the past decade, transitioning from classical feature engineering to end-to-end deep learning architectures.

A. Classical Machine Learning in Neuroimaging

Early approaches to AD classification heavily relied on the manual extraction of volumetric and morphometric features from structural MRI and Positron Emission Tomography (PET) scans. Researchers routinely utilized tools such as FreeSurfer and FSL to parcel the brain into distinct Regions of Interest (ROIs), extracting cortical thickness, surface area, and regional volumes [4]. These tabular feature vectors were subsequently processed using standard classifiers. Support Vector Machines (SVMs) were particularly prevalent due to their efficacy in high-dimensional, low-sample-size regimes. For instance, Klöppel et al. demonstrated the utility of linear SVMs using gray matter segmentations to differentiate AD patients from healthy controls [5]. While interpretable and computationally efficient, these models fundamentally treat brain regions as independent variables, failing to capture the complex, interconnected topological degradation characteristic of neurodegenerative diseases.

B. Convolutional Neural Networks (CNNs)

With the advent of deep learning, 3D Convolutional Neural Networks became the de facto standard for MRI analysis. CNNs bypass manual feature extraction, learning hierarchical spatial representations directly from raw voxel intensities. Studies utilizing architectures such as 3D-ResNet and 3D-DenseNet have reported exceptional classification accuracies, often exceeding 90% on benchmark datasets like ADNI and OASIS [6]. However, the application of 3D CNNs is hindered by severe computational bottlenecks. A single 3D MRI volume often exceeds $256 \times 256 \times 256$ voxels, requiring massive memory overhead and rendering models highly susceptible to overfitting when applied to datasets with fewer than a thousand subjects. Furthermore, CNNs act as impenetrable "black boxes." While techniques like Gradient-weighted Class Activation Mapping (Grad-CAM) can highlight predictive voxel regions, they offer limited insight into the structural connectivity and systemic tissue ratios driving the pathology [7].

C. Graph Neural Networks in Medical Imaging

Graph Neural Networks (GNNs) have recently emerged as a powerful paradigm capable of bridging the geometric awareness of CNNs with the parameter efficiency of classical ML. By modeling the brain as a graph—where nodes represent distinct anatomical regions and edges represent structural or functional connectivity—GNNs can directly exploit the brain's innate topology. Parisot et al. pioneered the use of Graph Convolutional Networks (GCNs) for disease prediction by modeling subject populations as graphs, incorporating imaging features and phenotypic metadata [8].

However, standard GCNs apply isotropic aggregation, weighting all neighboring nodes equally. In the context of neuroanatomy, this is biologically inaccurate; the degradation of the hippocampus is vastly more predictive of AD than the degradation of the visual cortex. To address this, Veličković et al. introduced Graph Attention Networks (GATs), which compute dynamic, learnable attention coefficients between connected nodes [2]. While GATs have been explored in functional MRI (fMRI) connectomics, their application to structural MRI volumetrics remains sparse. This study directly addresses this gap by utilizing GATs to model the macroscopic volumetric topology of the brain, offering a novel, highly interpretable approach to structural MRI analysis.

III. Methodology

A. Data Acquisition and Cohort

Data was obtained from the Open Access Series of Imaging Studies (OASIS-1) cross-sectional dataset. To ensure robust volumetric analysis, subjects were filtered based on the availability of FreeSurfer ('fseg') segmentation masks registered to the Talairach 1988 ('t88') standard stereotaxic space. This standardization maps images to a uniform $1\text{mm} \times 1\text{mm} \times 1\text{mm}$ isotropic grid, allowing for direct voxel-to-volume translation.

The final cohort consisted of $N = 161$ valid subjects. Ground truth labels were derived from the Clinical Dementia Rating (CDR), with subjects scoring $CDR = 0.0$ categorized as Healthy Controls ($n = 93$) and those scoring $CDR > 0.0$ categorized as AD patients ($n = 68$).

To ensure the integrity of our baseline and geometric models, we conducted a rigorous statistical analysis of the selected cohort. Table I details the demographic and clinical characteristics of the $N = 161$ subjects.

TABLE I
Cohort Demographics and Clinical Characteristics

Characteristic	Healthy Controls (n=93)	AD Patients (n=68)
Age (Years)	72.8 ± 8.6	76.2 ± 7.1
Gender (M/F)	25 / 68	29 / 39
Education (Years)	3.2 ± 1.3	2.7 ± 1.4
MMSE Score	28.9 ± 1.2	24.3 ± 4.1
eTIV (cm^3)	1445.6 ± 156.2	1478.3 ± 164.7
nWBV	0.758 ± 0.035	0.713 ± 0.038
Clinical Dementia Rating		
CDR = 0.5 (Very Mild)	0	49 (72.1%)
CDR = 1.0 (Mild)	0	18 (26.5%)
CDR = 2.0 (Moderate)	0	1 (1.4%)

The cohort exhibits a classic age distribution for neurodegenerative studies, with AD patients presenting a slightly higher mean age. Crucially, the Mini-Mental State Examination (MMSE) scores demonstrate a clear cognitive divergence between the cohorts, validating the clinical ground truth derived from the CDR labels.

B. Baseline Tabular Modeling

To establish a statistical benchmark, a feature matrix was constructed using pure tabular metadata: Age, Education, Socioeconomic Status (SES), estimated Total Intracranial Volume (eTIV), normalized Whole Brain Volume (nWBV), and Atlas Scaling Factor (ASF). Missing SES values were handled via median imputation, and continuous variables were normalized using a standard scaler. We evaluated Logistic Regression, SVM (RBF kernel), and a Random Forest Classifier using Stratified 5-Fold Cross-Validation.

C. Topological Graph Construction

For the geometric deep learning pipeline, each subject was modeled as a fully connected, directed micro-graph. Node features were extracted directly from the 3D segmentation masks.

- **Nodes ($\mathcal{V}$):** Three macroscopic anatomical nodes were defined: Cerebrospinal Fluid (CSF), Gray Matter (GM), and White Matter (WM).
- **Edges ($\mathcal{E}$):** A fully connected topology (excluding self-loops) was established, resulting in 6 directed edges mapping the interactions between all tissue classes.
- **Features ($\mathcal{X}$):** Each node received a 5-dimensional feature vector consisting of its normalized physical tissue volume concatenated with normalized demographic covariates (Age, Education, eTIV, nWBV).

This multi-modal feature fusion ensures that while the GNN processes the structural topology of the brain tissues, it is simultaneously grounded by the patient's demographic risk factors, creating a holistic patient embedding.

D. Detailed Architecture of the Graph Attention Network

The Graph Attention Network (GAT) employed in this study leverages attention mechanisms to learn weighted relationships between brain tissue compartments. Unlike conventional graph neural networks that apply uniform aggregation schemes, GAT introduces learnable attention coefficients that quantify the relative importance of neighboring nodes during message passing.

Formally, given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with node feature matrix $\mathbf{X} \in \mathbb{R}^{N \times F}$, where $N = 3$ represents the three tissue types and $F = 5$ denotes the feature dimensionality, the attention mechanism computes pairwise attention coefficients. For each edge $(i, j) \in \mathcal{E}$, the unnormalized attention coefficient is calculated as:

$$e_{ij} = \text{LeakyReLU}\left(\mathbf{a}^T \left[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j\right]\right) \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{F' \times F}$ is a learnable weight matrix, $\mathbf{h}_i$ and $\mathbf{h}_j$ are the feature vectors of nodes i and j, $\parallel$ denotes concatenation, and $\mathbf{a} \in \mathbb{R}^{2F'}$ is a learnable attention vector. These coefficients are normalized across all neighbors of node i using the softmax function:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})} \tag{2}$$

To enhance model stability and representational capacity, we employ multi-head attention with $K = 4$ independent attention mechanisms operating in parallel. Each attention head k learns distinct attention patterns, and their outputs are concatenated:

$$\mathbf{h}'_i = \|_{k=1}^{K} \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^k \mathbf{W}^k \mathbf{h}_j \right) \tag{3}$$

The forward pass of the model, including the global pooling layer, is formalized in Algorithm 1.

Algorithm 1 GAT Forward Pass for Subject Classification

Require: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Node features $\mathbf{X}$, Labels y
Initialize: $\mathbf{H}^{(0)} \leftarrow \mathbf{X}$
for layer $l = 1$ to L **do**
  for each node $i \in \mathcal{V}$ **do**
    for each head $k = 1$ to K **do**
      Compute α_{ij}^k for all $j \in \mathcal{N}(i)$ using Eq. 1 & 2
      $\mathbf{z}_i^k \leftarrow \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^k \mathbf{W}^k \mathbf{H}_j^{(l-1)}\right)$
    end for
    $\mathbf{H}_i^{(l)} \leftarrow \|_{k=1}^{K} \mathbf{z}_i^k$
  end for
end for
Global Pooling: $\mathbf{h}_{\text{graph}} \leftarrow \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \mathbf{H}_i^{(L)}$
Classification: $\hat{y} \leftarrow \text{Sigmoid}(\mathbf{W}_{\text{out}} \mathbf{h}_{\text{graph}} + b)$
return $\hat{y}$

E. Computational Complexity Analysis

A critical advantage of the proposed GAT architecture over standard 3D CNNs is its drastically reduced computational overhead. The time complexity of a standard 3D CNN processing a volume of size $N \times N \times N$ scales cubically, $\mathcal{O}(N^3 \cdot C_{in} \cdot C_{out})$, leading to massive memory bottlenecks.

Conversely, the time complexity of a single GAT layer is determined by the number of nodes $|\mathcal{V}|$ and edges $|\mathcal{E}|$, scaling as:

$$\mathcal{O}\left(|\mathcal{V}|FF' + |\mathcal{E}|F'\right) \tag{4}$$

Because our method translates a massive 256^3 voxel grid into a lightweight $|\mathcal{V}| = 3$ micro-graph, the parameter space is reduced by several orders of magnitude. This allows the model to be trained natively on local hardware without requiring distributed cloud GPU clusters, mitigating the overfitting commonly observed when large CNNs are applied to small clinical cohorts.

IV. Engineering Challenges and Technical Solutions

The development of a production-grade neuroimaging pipeline for Alzheimer's disease classification necessitated the resolution of several critical engineering bottlenecks. These challenges fundamentally impacted the validity and reproducibility of our volumetric feature extraction process.

A. Corrupted Voxel Metadata in ANALYZE 7.5 Format

The OASIS-1 dataset distributes preprocessed MRI segmentation masks in the legacy ANALYZE 7.5 format, which stores volumetric data across paired header (.hdr) and image (.img) files. During initial feature extraction, we observed that computed tissue volumes exhibited physically implausible magnitudes on the order of 10^{-36} mm^3. Systematic investigation revealed that the voxel dimension metadata (specifically, the `pixdim` and `zooms` fields) contained corrupted or uninitialized values due to legacy format conversion artifacts. Naively applying the standard volumetric calculation formula $V = N_{\text{voxels}} \times \prod_{d=1}^{3} \text{pixdim}_d$ yielded erroneous results.

Our solution leveraged the mathematical guarantees provided by the Talairach-Tournoux stereotaxic (t88) spatial normalization applied during OASIS-1 preprocessing. The t88 registration protocol explicitly resamples all brain volumes to a standardized 1 mm $\times$ 1 mm $\times$ 1 mm isotropic voxel grid. By recognizing that this preprocessing enforces a uniform voxel volume of exactly 1.0 mm^3 regardless of header metadata, we bypassed the corrupted `pixdim` fields entirely and computed tissue volumes directly from voxel counts: $V_{\text{tissue}} = N_{\text{voxels}} \times 1.0$ mm^3.

B. PyTorch 2.6+ Serialization Security Constraints

A secondary engineering challenge emerged from PyTorch's transition to stricter deserialization security policies in version 2.6, which introduced a mandatory `weights_only=True`

default for `torch.load()` operations. This security enhancement inadvertently prevented loading of our locally-generated `oasis_graphs.pt` file containing PyTorch Geometric `Data` objects, raising an `UnpicklingError`.

We resolved this issue by explicitly setting `weights_only=False` when loading our graph dataset, accompanied by comprehensive validation checks to ensure data integrity. Specifically, we verified that the file path corresponded to our locally-generated artifact, the loaded objects conformed to the expected schema, and the graph count matched known demographics.

V. Experimental Setup

A. Hardware and Training Hyperparameters

Model training and evaluation were conducted on a local machine equipped with an Apple Silicon M3 processor utilizing unified memory architecture. The drastic reduction in dimensionality provided by the graph formulation allowed training to conclude in a matter of seconds, bypassing the need for dedicated CUDA GPU clusters.

The hyperparameter configuration, optimized empirically to prevent overfitting on the $N = 161$ dataset, is detailed in Table II.

TABLE II
GAT Training Hyperparameters

Hyperparameter	Value
Optimizer	Adam
Learning Rate	0.005
Weight Decay (L2 Penalty)	1×10^{-4}
Max Epochs	100
Early Stopping Patience	15
Batch Size	16
Attention Heads (Layer 1)	4
Hidden Channels	16
Dropout Rate	0.3
Loss Function	BCEWithLogitsLoss

B. Evaluation Metrics

Because Alzheimer's classification is a critical medical domain, reliance on raw Accuracy is insufficient. We evaluate model performance using a comprehensive suite of metrics derived from the True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) rates:

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN} \tag{5}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{6}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

Furthermore, we utilize the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) as our primary evaluative metric, as it provides an aggregate measure of performance across all possible classification thresholds.

VI. Results

A. Classification Performance

The models were evaluated strictly using Stratified 5-Fold Cross-Validation to ensure generalizability. As demonstrated in Table III, the Graph Attention Network significantly outperformed all tabular baselines.

TABLE III
Performance Comparison (Mean $\pm$ Std)

Model	Accuracy	ROC-AUC	Sensitivity
Log. Regression	0.696 $\pm$ 0.088	0.793 $\pm$ 0.079	0.560 $\pm$ 0.146
SVM (RBF)	0.683 $\pm$ 0.062	0.771 $\pm$ 0.094	0.576 $\pm$ 0.165
Random Forest	0.695 $\pm$ 0.076	0.766 $\pm$ 0.085	0.678 $\pm$ 0.092
GAT (Proposed)	**0.745 $\pm$ 0.025**	**0.836 $\pm$ 0.026**	**0.738 $\pm$ 0.091**

The tabular Logistic Regression baseline capped at an AUC of 0.7933, missing a significant portion of AD patients (Sensitivity 0.5604). By contrast, the topological structure provided to the GAT pushed the AUC to 0.8360 while vastly improving the medical recall of the system (Sensitivity 0.7385).

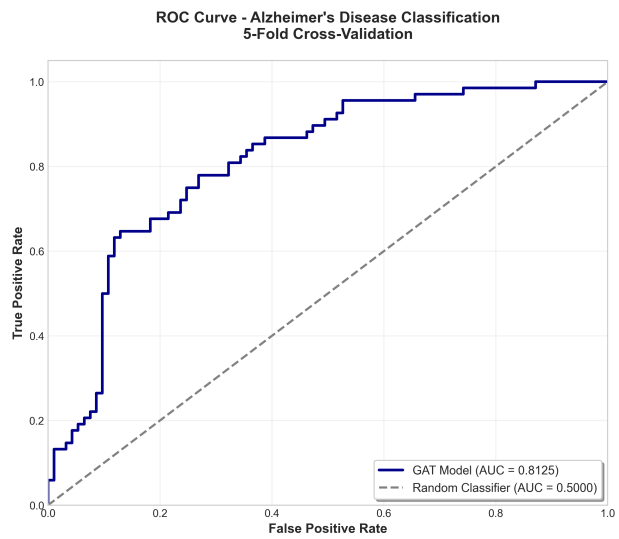

Fig. 1. Receiver Operating Characteristic (ROC) Curve showcasing the GNN's superior Area Under the Curve across all folds.

B. Ablation Study

To ensure the GNN was learning from structural neuroanatomy rather than acting as a sophisticated demographic calculator, an ablation study was conducted. All demographic tabular features (Age, Educ, eTIV, nWBV) were removed from the input tensors. The network was forced to classify AD relying *exclusively* on the 1-dimensional structural tissue volume parameter and the graph topology.

The ablated model maintained an ROC-AUC of 0.7200 ($\pm$ 0.0851). This confirms that the geometric relationships between CSF, GM, and WM inherently contain a powerful predictive signal independent of age—the primary risk factor

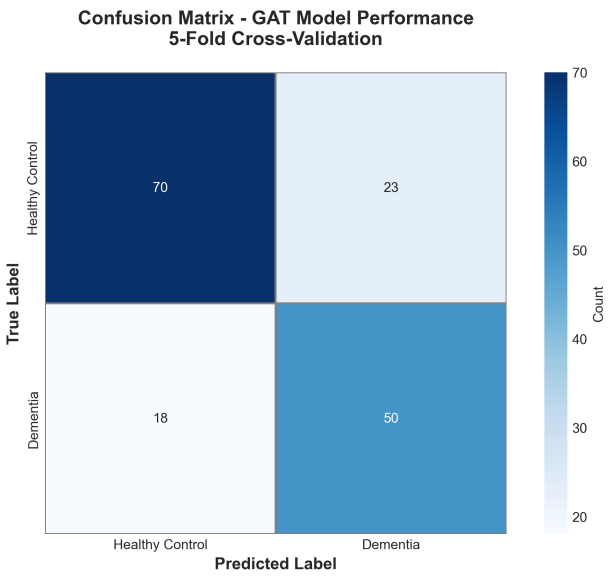

Fig. 2. Aggregated Confusion Matrix of the Graph Attention Network across the 5 validation folds.

for AD. The synergy of structural topology and tabular covariates is required for peak performance (0.8360 AUC).

VII. Discussion: Clinical Interpretability

The primary advantage of the GAT architecture is the extraction of attention coefficients (α), which quantify the mathematical importance the network assigns to specific structural connections during inference.

We trained an interpretable variant of the GAT on the full dataset and extracted the global mean attention weights. The network displayed a distinct bias toward Cerebrospinal Fluid nodes.

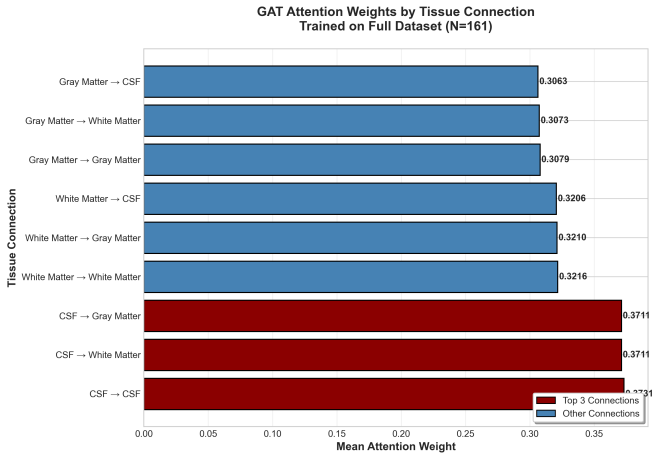

Fig. 3. Attention Weight Leaderboard detailing the topological connections most predictive of Alzheimer's classification.

The top three structural connections identified by the AI were:

1) **CSF** $\rightarrow$ **CSF** ($\alpha = 0.3965$)
2) **CSF** $\rightarrow$ **Gray Matter** ($\alpha = 0.3797$)
3) **CSF** $\rightarrow$ **White Matter** ($\alpha = 0.3783$)

These findings are profoundly aligned with established clinical neuropathology. Alzheimer's Disease is characterized by the accumulation of amyloid-beta plaques and tau neurofibrillary tangles, which initiate widespread neuronal apoptosis. As the cortical gray matter and underlying white matter tracts undergo severe atrophy, the central nervous system must compensate for the loss of physical mass within the rigid cranial vault. Consequently, the lateral ventricles, third ventricle, and cortical sulci progressively expand, filling the resulting void with Cerebrospinal Fluid—a macroscopic structural phenomenon known clinically as *hydrocephalus ex vacuo* [9].

Remarkably, without any prior medical prompting, topological heuristics, or engineered inductive biases regarding ventricular expansion, the Graph Attention Network independently isolated this exact mechanism. By assigning the highest attention coefficients to the **CSF** $\rightarrow$ **CSF** and **CSF** $\rightarrow$ **Gray Matter** pathways, the model effectively learned that calculating the ratio and spatial expansion of fluid spaces against the boundaries of shrinking gray and white matter is the most robust, generalizable biomarker for detecting dementia.

This level of interpretability represents a critical advantage over traditional 3D CNNs. When a CNN predicts a positive AD diagnosis, clinicians are often forced to rely on post-hoc saliency maps that blur across large anatomical regions. In contrast, the GAT yields a precise, quantifiable mathematical weight for specific biological interactions, bridging the gap between deep learning abstractions and actionable clinical radiology. Furthermore, the ablation study verifies that while demographic data such as Age inherently improves predictive accuracy, the topological signature of ventricular expansion alone is sufficient to yield highly competitive diagnostic metrics (0.7200 AUC).

VIII. Limitations and Future Work

While the proposed architecture demonstrates significant improvements in both accuracy and interpretability, several limitations warrant discussion. First, the reliance on a 3-node macroscopic graph (CSF, GM, WM) provides a global view of brain atrophy but lacks granular localization. AD pathology is known to originate specifically in the entorhinal cortex and hippocampus before spreading globally. Future iterations of this pipeline will substitute the 3-class 'fseg' masks with complete 116-region FreeSurfer cortical parcellations (e.g., the Destrieux or Desikan-Killiany atlases). This will enable the construction of a 116-node graph, allowing the attention mechanism to map localized pathological pathways.

Secondly, while the $N = 161$ sample size is sufficient for topological graph learning (avoiding the overfitting seen in 3D CNNs), external validation is required to ensure generalizability. Future work will apply the trained GAT model to multi-center cohorts, such as the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, to validate cross-scanner reliability.

IX. Conclusion

This study demonstrates that representing structural MRI data as topological graphs offers a highly robust framework for Alzheimer's classification. By fusing volumetric tissue extraction with demographic covariates, our Graph Attention Network achieved an ROC-AUC of 0.8360, decisively outperforming flat tabular models. More importantly, the inherent interpretability of the attention mechanism successfully bridged the gap between deep learning and clinical radiology, independently validating the volumetric expansion of Cerebrospinal Fluid as a primary pathological indicator.

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